Project Plan

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Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

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https://github.com/ese-ada-lovelace-2025/irp-th25

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Abstract

Wildfires are destructive and a rapidly evolving hazard. With climate change, they become increasingly severe and frequent. This increases the need for a higher-resolution wildfire data for better monitoring and risk assessment. Geostationary satellites (GEO) have high-temporal resolution, but lacks the resolution needed for a detailed wildfire risk assessment. While low orbit satellites (LEO) provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing downscaling models have demonstrated their effectiveness in producing high-resolution remote sensing data. But many still suffers from artefacts, blurriness, and hallucinations. This project aims to achieve a high-resolution spatial downscaling of wildfire data by integrating additional physical covariates to the model, such as topography and landcover to mitigate physically implausible outputs. The project compares direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-fidelity, high-resolution wildfire data. The project compares diffusion (EDiffSR), attention U-Net (HAUNet), and transformer-based (ESTNet) downscaling models.

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1 Problem Description and Motivation

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). This causes economic losses for the insurance industry. For instance, The January 2025 Los Angeles County wildfires set an all-time record, causing $40 billion in insured losses alone from a single event (Dwyer and Nam, 2026). This project, conducted with Artio, aims to address this gap. By providing highly accurate risk assessment models, this research seeks to give better confidence and enable better risk assessment.

High-spatial and high-temporal resolution wildfire data is highly desired in risk assessment. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate physically implausible outputs. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net, a transformer-based model, and a diffusion-based model.

2 Literature Review

2.1 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns and variability in remote sensing data. In recent years, DL methods have shown promising results that outpaces traditional approaches (Sdraka et al., 2022). In particular, Convolutional Neural Networks (CNNs) and U-Net architectures have been widely used for their ability to capture spatial features. Generative Adversarial Networks (GANs) have also been used for their ability to generate sharp images. Recently, Transformer and Diffusion-based approaches have been proposed for downscaling. Transformers use attention mechanisms to capture long-range dependencies (J. Wang et al., 2025), while Diffusion models shows good generative ability and fidelity (Y. Wang et al., 2023).

However, many of these approaches have limitations, particularly in preserving fine-grained textures and geometric structures (Qi et al., 2026). This is problematic for wildfire data, where the spatial patterns and textures are crucial for accurate risk assessment. To address these limitations, there is a need for domain-specific architectures that can better capture the characteristics of wildfire data.

Furthermore, existing literature remains reliant on direct downscaling methods across large resolution-gaps. Some studies have explored cascaded approaches, such as AMCN-Geoi-DBN from Jing et al., 2022, and CLIT from Chen et al., 2023, but there are currently no research from the literature investigating cascaded downscaling in the context of wildfires. The limitations and benefits of cascaded downscaling is also largely unexplored. To bridge this gap, this project

introduces a novel, domain-specific framework, and explores the benefits and limitations of cascaded downscaling.

2.2 Selected Downscaling Architectures for Comparison

2.2.1 HAUNet

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses hybrid attention to capture both local fine-grained detail and broader contextual structure across multiple scales. This allows it to capture local extreme hotspots in near-zero background, which standard U-nets might smooth out. Therefore, HAUNet will be selected as one of the architectures for comparison, making it a strong baseline for capturing spatially heterogeneous patterns in wildfire data.

2.2.2 ESTNet

ESTNet (Kang et al., 2024) is a transformer-based architecture that uses efficient spatial-temporal attention to capture long-range dependencies in the data. It is based on Swin-Transformers with group attention. It uses multi-window self-attention for better efficiency without sacrificing reconstruction quality. For transformer-based approach, ESTNet will be selected for comparison. Their ability in capturing long-range dependencies, makes them a suitable candidate for downscaling wildfire data, which has complex spatial patterns. Comparing ESTNet specifically is preferred over alternatives such as SwinIR (Liang et al., 2021) because their efficiency could potentially make up for the computational cost in cascaded approaches.

2.2.3 EDiffSR

EDiffSR (Xiao et al., 2023) is a conditional diffusion-based model. It uses simplified channel attention for better noise prediction quality, and Conditional Prior Enhancement Module (CPEM) to extract the most valuable information from the LR input. EDiffSR is selected as the diffusion-based model for comparison, because it addresses the problem of one-to-many mapping by effectively leveraging prior information and promotes the condition with more informative input. This makes it a suitable candidate for this project, which has a significant resolution gap and complex spatial patterns.

3 Methodology

The architectures (HAUNet, ESTNet, EDiffSR) for comparison is tentatively selected based on their performance in downscaling, and their suitability for the specific requirements of the wildfire data. The reasons for selecting each architecture are outlined in 2.2. The final selection of architectures and datasets may be adjusted as the project progresses.

3.1 Datasets

3.1.1 Input Datasets

The input data are:

- GOES-17 channel 7 (IR shortwave) of Level 1B (L1B) Radiances product (ABI-L1B-Rad). Has spatial resolution of 2km at every 5 mins. This channel is selected because it is commonly used to identify hot regions (Badhan et al., 2024). This will be the main coarse input.
- Copernicus DEM GLO-30: Global 30m Digital Elevation Model with 30m resolution (European Space Agency and Airbus, 2022). DEM band will be used.

- ESA WorldCover 10m v100 landcover dataset with 10m resolution. This dataset have 11 land cover classes, including tree cover, shrubland, grassland, and permanent water bodies (Zanaga et al., 2021).
- Other potential covariates include weather data, such as wind speed and direction, temperature, and humidity. The inclusion of these covariates will depend on their availability, relevance and time constraints.

3.1.2 Target Datasets

The target data are:

- VIIRS (S-NPP) Band Active Fire with 375 m spatial resolution (Earth Science Data Systems, 2025) will be used as the target data for the 2km to 375m stage downscaling. The I-4 channel brightness temperature (3.55-3.93 µm) band will be used.
- Landsat-8 active fire dataset includes data for fire scenes taken from August and September 2020 with 30m resolution (de Almeida Pereira et al., 2021). This will be the final target HR target data. Fire-sensitive SWIR and thermal bands will be used.

3.2 Experimental Design

The project will evaluate the performance of the selected architectures in both direct and cascaded downscaling pipelines. The direct pipeline will train the model to directly downscale from 2km to 30m resolution. The cascaded pipeline will involve two stages: the first stage downscales from 2km to 375m (trained on GOES and VIIRS data), and the second stage will further downscale from 375m to 30m (trained on VIIRS and Landsat-8 data). The final 30m resolution output from both pipelines will be compared against the Landsat-8 active fire dataset for evaluation.

Table 1: Summary of Experimental Configurations

Experiment	Architecture	Pipeline	Input Resolution	Output Resolution	Model Type
E1	HAUNet	Direct	2km (GOES-17)	30m (Landsat-8)	Hybrid Attention U-Net
E2	HAUNet	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Hybrid Attention U-Net
E3	EDiffSR	Direct	2km (GOES-17)	30m (Landsat-8)	Diffusion
E4	EDiffSR	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Diffusion
E5	ESTNet	Direct	2km (GOES-17)	30m (Landsat-8)	Transformer
E6	ESTNet	Cascaded	Stage 1: 2km → 375m Stage 2: 375m → 30m	30m (Landsat-8)	Transformer

3.3 Evaluation Metrics

The metrics for evaluation will include:

- Intersection over Union (IoU): Measures the overlap between the predicted fire regions and the actual fire regions in the target data. Higher IoU indicates better performance in capturing the spatial extent of fires.
- Intersection's Peak Signal-to-Noise Ratio (IPSNR): Quantifies the similarity of values in the intersection of the predicted and target fire regions. This is chosen over PSNR because fires only occupy small portion of the image, and PSNR would predict background to be similar (Badhan et al., 2024).

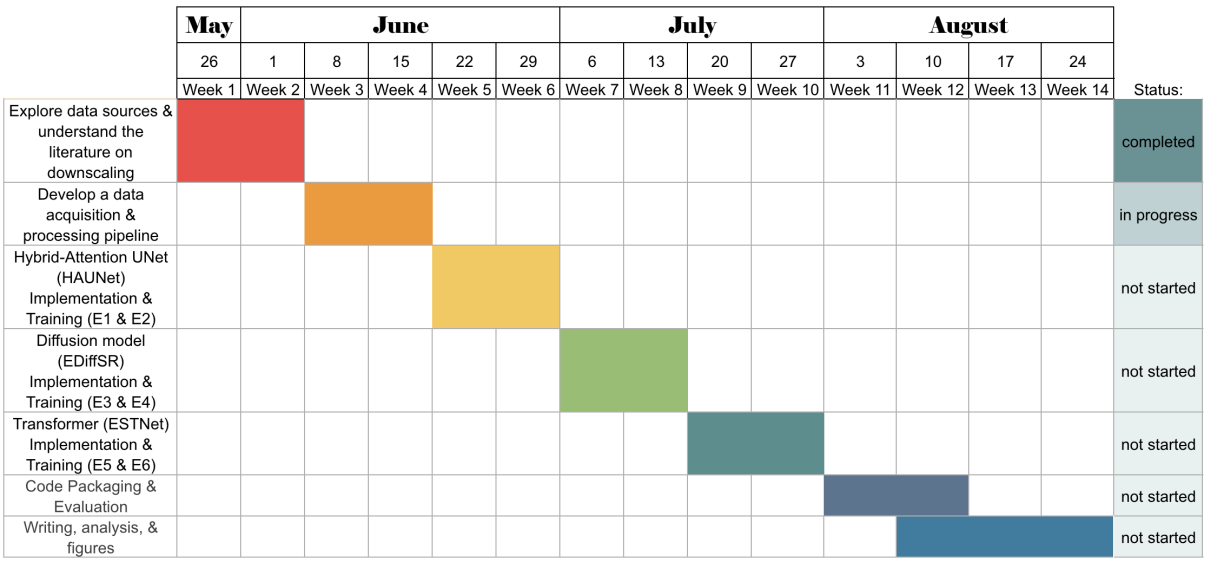


Figure 1: Gantt Chart of Project Timeline and Milestones

- F1 Score / Dice Coefficient: Evaluate similarity or overlap between the predicted and ground truth. They are often used in image segmentation.

4 Progress to Date

The project is currently in the early stages of data collection and preprocessing. The data acquisition and processing pipeline is being developed. The key milestones for the project is shown in Fig. 1.

5 Future Plans

The next steps for the project include:

1. Developing the data preprocessing pipeline to prepare the input and target datasets for training. This includes aligning the datasets spatially and temporally, and integrating the additional covariates.
2. Implementing the selected architectures and training them on the respective datasets for both direct and cascaded pipelines.
3. Evaluating the performance of the models using the defined metrics and comparing the results between direct and cascaded pipelines.
4. Conducting a detailed analysis of the results to understand the strengths and limitations of each approach, and identifying potential areas for improvement.
5. Preparing the final report and presentation summarising the findings and insights from the project.

References

Badhan, M., Shamsaei, K., Ebrahimian, H., Bebis, G., Lareau, N. P., & Rowell, E. (2024). Deep Learning Approach to Improve Spatial Resolution of GOES-17 Wildfire Boundaries Using VIIRS Satellite Data. *Remote Sensing*, *16*(4), 715. https://doi.org/10.3390/rs16040715

Chen, H.-W., Xu, Y.-S., Hong, M.-F., Tsai, Y.-M., Kuo, H.-K., & Lee, C.-Y. (2023). Cascaded Local Implicit Transformer for Arbitrary-Scale Super-Resolution, 18257–18267. Retrieved June 7, 2026, from https://openaccess.thecvf.com/content/CVPR2023/html/Chen_Cascaded_Local_Implicit_Transformer_for_Arbitrary-Scale_Super-Resolution_CVPR_2023_paper.html

de Almeida Pereira, G. H., Fusioka, A. M., Nassu, B. T., & Minetto, R. (2021). Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study. *ISPRS Journal of Photogrammetry and Remote Sensing*, *178*, 171–186. https://doi.org/10.1016/j.isprsjprs.2021.06.002

Dwyer, O., & Nam, H. W. (2026, March 31). *How wildfires and storms drove insurance losses in 2025 – in three charts*. Carbon Brief. Retrieved June 10, 2026, from https://www.carbonbrief.org/how-wildfires-and-storms-drove-insurance-losses-in-2025-in-three-charts/

Earth Science Data Systems, N. A. S. A. (2025, May 27). *VIIRS (S-NPP) I Band 375 m Active Fire Product NRT (Vector data) | NASA Earthdata*. Earth Science Data Systems, NASA. Retrieved June 8, 2026, from https://www.earthdata.nasa.gov/data/catalog/lancemodis-vnp14imgtdl-nrt-2

European Space Agency & Airbus. (2022). *Copernicus DEM*. https://doi.org/10.5270/ESA-c5d3d65

Jing, Y., Lin, L., Li, X., Li, T., & Shen, H. (2022). Cascaded Downscaling–Calibration Networks for Satellite Precipitation Estimation. *IEEE Geoscience and Remote Sensing Letters*, *19*, 1–5. https://doi.org/10.1109/LGRS.2022.3214083

Jones, M. W., Smith, A. J. P., Betts, R., Canadell, J. G., Prentice, I. C., & Quéré, C. L. (2020). Climate Change Increases the Risk of Wildfires.

Kang, X., Duan, P., Li, J., & Li, S. (2024). Efficient Swin Transformer for Remote Sensing Image Super-Resolution. *IEEE Transactions on Image Processing*, *33*, 6367–6379. https://doi.org/10.1109/TIP.2024.3489228

Lambert, J., & Turner, G. (2024, November 14). *Investment Trends and Outcomes in the Global Carbon-Credit Market-2024 | MSCI*. Retrieved June 5, 2026, from https://www.msci.com/research-and-insights/paper/investment-trends-and-outcomes-in-the-global-carbon-credit-market-2024

Liang, J., Cao, J., Sun, G., Zhang, K., Gool, L. V., & Timofte, R. (2021, August 23). *SwinIR: Image Restoration Using Swin Transformer*. arXiv: 2108.10257 [eess.IV]. https://doi.org/10.48550/arXiv.2108.10257

Qi, Y., Lou, M., Liu, Y., Li, L., Yang, Z., & Nie, W. (2026). Advancing image super-resolution techniques in remote sensing: A comprehensive survey. *ISPRS Journal of Photogrammetry and Remote Sensing*, *231*, 68–100. https://doi.org/10.1016/j.isprsjprs.2025.10.024

Sdraka, M., Papoutsis, I., Psomas, B., Vlachos, K., Ioannidis, K., Karantzalos, K., Gialampoukidis, I., & Vrochidis, S. (2022). Deep Learning for Downscaling Remote Sensing Images: Fusion and super-resolution. *IEEE Geoscience and Remote Sensing Magazine*, *10*(3), 202–255. https://doi.org/10.1109/MGRS.2022.3171836

Wang, J., Chen, M., Karaev, N., Vedaldi, A., Rupprecht, C., & Novotny, D. (2025, March 14). *VGGT: Visual Geometry Grounded Transformer*. arXiv: 2503.11651 [cs.CV]. https://doi.org/10.48550/arXiv.2503.11651

Wang, J., Wang, B., Wang, X., Zhao, Y., & Long, T. (2023). Hybrid Attention-Based U-Shaped Network for Remote Sensing Image Super-Resolution. *IEEE Transactions on Geoscience and Remote Sensing*, *61*, 1–15. https://doi.org/10.1109/TGRS.2023.3283769

Wang, Y., Yang, W., Chen, X., Wang, Y., Guo, L., Chau, L.-P., Liu, Z., Qiao, Y., Kot, A. C., & Wen, B. (2023, November 23). *SinSR: Diffusion-Based Image Super-Resolution in a Single Step*. arXiv: 2311.14760 [cs.CV]. https://doi.org/10.48550/arXiv.2311.14760

Wang, Z., Chen, J., & Hoi, S. C. H. (2020, February 8). *Deep Learning for Image Super-resolution: A Survey*. arXiv: 1902.06068 [cs.CV]. https://doi.org/10.48550/arXiv.1902.06068

Xiao, Y., Yuan, Q., Jiang, K., He, J., Jin, X., & Zhang, L. (2023, October 30). *EDiffSR: An Efficient Diffusion Probabilistic Model for Remote Sensing Image Super-Resolution*. arXiv: 2310.19288 [eess.IV]. https://doi.org/10.48550/arXiv.2310.19288

Zanaga, D., Van De Kerchove, R., De Keersmaecker, W., Souverijns, N., Brockmann, C., Quast, R., Wevers, J., Grosu, A., Paccini, A., Vergnaud, S., Cartus, O., Santoro, M., Fritz, S., Georgieva, I., Lesiv, M., Carter, S., Herold, M., Li, L., Tsendbazar, N.-E., ... Arino, O. (2021, October 20). *ESA WorldCover 10 m 2020 v100*. Zenodo. https://doi.org/10.5281/zenodo.5571936